

Stony Brook University  
Alejandro Melo Ponce

Spring 2019

### ASSIGNMENT 1: Math Review

**Due: Monday, February 11 in class.**

Instructions: Show all your work or points will be deducted.

1. Graph the following functions. Be as precise as you can, i.e. indicate intercepts, asymptotes, critical points and inflection points, if any.

(a)  $y = -2x + 4$

(b)  $y = \frac{2}{x}$

(c)  $y = 16x - x^2$

(d)  $y = 4x - \frac{1}{x^2}$

2. Find  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  for the functions below, where  $\alpha$  and  $\beta$  are positive constants.

(a)  $y = \alpha x^2 + \beta x$

(b)  $y = \beta \ln(\alpha x)$

(c)  $y = \alpha e^{\beta x}$

(d)  $y = \beta z^2$  and  $z = \alpha x + 5$

(e)  $y = \frac{\alpha x - \beta}{x - \alpha}$

3. Suppose a firm's total revenues depend on the amount produced ( $q$ ) according to the function

$$R = 70q - q^2.$$

Total costs also depend on  $q$ :

$$C = q^2 + 30q + 100.$$

- (a) What level of output should the firm produce to maximize profits ( $R - C$ )? What will profits be?
- (b) Show that the second-order conditions for a maximum are satisfied at the output level found in part (a).
- (c) Does the solution that you calculated obey the "**Marginal revenue equals Marginal cost**" rule? Explain.